Chin J Integr Med 2018 Jan;24(1):32-39

Chinese Journal of Integrative Medicine

Available online at link.springer.com/journal/11655
Journal homepage: www.cjim.cn/zxyjhen/zxyjhen/ch/index.aspx
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Original Article

Protective Effect of Hydroxysafflor Yellow A on Bleomycin-Induced Pulmonary Inflammation and Fibrosis in Rats*

JIN Ming, WANG Lin, WU Yan, ZANG Bao-xia, and TAN Li

ABSTRACT **Objective**: To observe the effect of hydroxysafflor yellow A (HSYA), an active ingredient of a traditional Chinese herbal medicine *Carthamus tinctorius* L., on lung inflammation and pulmonary fibrosis induced by bleomycin (BLM) in rats. **Methods**: Animals were divided into 6 groups including normal group, model group, three HSYA groups and dexamethasone (DXM) group. Three doses of HSYA (35.6, 53.3, and 80.0 $mg\cdot kg^{-1}\cdot day^{-1}$) were intraperitoneally (i.p.) injected in rats for 3 weeks after BLM administration and DXM was used as the positive control (n=8 or 12). Arterial blood gas was assayed and morphological changes were observed. Lung mRNA expressions of tumor necrosis factor (TNF)-α, interleukin (IL)-1β, IL-6 and some cytokines in lung tissue were detected by real-time polymerase chain reaction. Nuclear factor-κB p65 or α-smooth muscle actin (α-SMA) protein distribution in rat lung tissue was observed by immunohistochemistry. **Results**: On the 7th day after BLM administration, lung tissue showed serious inflammation. Treatment with HSYA or DXM ameliorated lung inflammation. After treatment with HSYA or DXM, oxygen partial pressure (PaO_2) increased (HSYA 80.0 $mg\cdot kg^{-1}$, P<0.01) and CO_2 partial pressure ($PaCO_2$) decreased (HSYA 53.3, 80.0 $mg\cdot kg^{-1}$, P<0.05). Moreover, the mRNA expression of TNF-α, IL-1β, and IL-6; and the number of NF-κB p65 positive cells was lower in HSYA 53.3 and 80.0 $mg\cdot kg^{-1}$ groups than those in the model group (all P<0.05). Twenty-one days after BLM administration, HSYA or DXM treatment ameliorated fibrosis, increased PaO_2 (HSYA 53.3, 80.0 $mg\cdot kg^{-1}$, P<0.01), and decreased $PaCO_2$ (53.3 and 80.0 $mg\cdot kg^{-1}$, P<0.05). Further, the mRNA expression of TGF-β1, α-SMA, and collagen Ⅰ as well as the number of α-SMA positive cells increased in the model group and HSYA can attenuate these changes (53.3, 80.0 $mg\cdot kg^{-1}$, P<0.05). Hematoxylin and eosin and Masson's trichrome staining indicated that the fibrosis and collagen deposition were ameliorated in HSYA groups (53.3, 80.0 $mg\cdot kg^{-1}$, P<0.05). **Conclusion**: HSYA could alleviate acute lung inflammation and chronic pulmonary fibrosis induced by BLM in rats.

KEYWORDS hydroxysafflor yellow A, pulmonary fibrosis, pulmonary inflammation, nuclear factor-κB p65, α-smooth muscle actin, Chinese medicine

Carthamus tinctorius L. (safflower) has been used to treat blood stasis as a traditional Chinese herbal medicine for thousands of years. Phytochemical research showed that safflor yellow (SY) is the effective part of safflower and hydroxysafflor yellow A (HSYA) is the main active ingredient of SY.(1) HSYA can inhibit carbon tetrachloride (CCl_4)-induced liver fibrosis,(2) but there are no studies on its pharmacological effects on pulmonary fibrosis. Bleomycin (BLM) is often used to induce pulmonary fibrosis *in vivo*.(3) In rodents, lung injury induced by intratracheal delivery of BLM is characterized by two sequential pathological phases: early acute lung inflammatory injury, and late pulmonary fibrosis.(4) Previous studies have shown that the transition between inflammation and fibrotic remodeling occurred approximately 9 days after BLM administration. Proinflammatory cytokines such as interleukin (IL)-1α, IL-1β, IL-6, and interferon (IFN)-γ were significantly elevated until day 9. This sustained damage leads to the initiation fibrosis, which is observed from about day 9 to day 21.(5) Inflammation is the initial response to various injuries.

*Supported by Traditional Chinese Medicine Development Foundation of Beijing (No. JJ-2009-22), and Natural Science Foundation of Beijing (No. 7132047)

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DOI: https://doi.org/10.1007/s11655-017-2094-z

Activated inflammatory cells such as macrophages and neutrophils accumulate, which release cytokines. Cytokines are important regulators of the inflammatory and fibrotic process.[6] Our previous study found that HSYA could alleviate acute lung injury induced by administration of lipopolysaccharide in mice.[7] Because the BLM model reproduces acute lung injury in the early phase, and pulmonary fibrosis in the late phase, we aimed to investigate the potential effects of HSYA on early inflammation and late fibrosis in this model. HSYA can inhibit BLM induced mice early inflammation[8] and SY injection can alleviate BLM induced rat pulmonary fibrosis,[9] but there was no report about the inhibition effect of HSYA against BLM-induced rat pulmonary fibrosis especially at the dose of inhibiting early inflammation injury. In this paper we investigated the potential of HSYA to attenuate BLM-induced rat early pulmonary inflammation and late fibrosis at the same dose, in order to find new medicine to treat pulmonary fibrosis.

METHODS

Materials and Animals

Carthamus tinctorius L. was provided by Huahuikaide Pharmaceutical Co., Ltd. (Shanxi, China), and was grown in Tacheng, Xinjiang Uygur autonomous region, China. Safflower was identified by Prof. LI Jia-shi (Beijing University of Chinese Medicine). HSYA was isolated and purified by macroporous resin-gel column chromatography from the aqueous extracts of *Carthamus tinctorius* L. as previously described.[10] The purity of HSYA was above 95% as determined by high-performance liquid chromatography (HPLC). HSYA was dissolved in sterile 0.9% NaCl solution for subsequent use.

BLM was from Nippon Kayaku Co., Ltd. (Tokyo, Japan). Dexamethasone (DXM) injection, used as a positive control drug, was from Jinhui Pharmaceutical Company (Jiaozuo, Henan, China). Heparin sodium salt was from Sinopharm Chemical Reagent Beijing Co., Ltd. (Beijing, China). Rabbit anti-nuclear factor-κB (NF-κB) p65 and α-smooth muscle actin (α-SMA) multicolony antibodies were from Santa Cruz (Dallas, TX, USA). Horseradish peroxidase-conjugated sheep anti-rabbit antibody (IgG) was from Beyotime Institute of Biotechnology (Jiangsu, China). TRIzol reagent was produced by Invitrogen Co. (Carlsbad, CA, USA.). SYBR® Premix Ex Taq™ (Perfect Real Time) was from Takara Bio Inc. (Otsu, Shiga, Japan). HPLC-grade acetonitrile was produced by Fisher Scientific Co., Ltd. (Fair Lawn, NJ, USA). All the other reagents were analytical grade and were produced in China.

Specific-pathogen-free mature male Wistar rats weighing 230–250 g were purchased from Beijing Vital River Experimental Animals Technology Ltd. (Beijing, China, Certification No. 0196503). Animals were maintained in the Animal Department of the Anzhen Hospital with controlled temperature (23 ± 2 ℃) and humidity (60% ± 10%), under a 12-h light/dark cycle. All animal experiments were performed in compliance with the Beijing Laboratory Animal Management Ordinance and the experimental procedure was approved by the administrative commissioner of the Ethic Committee of Capital Medical University.

Treatment and Grouping

Rats were stratified and randomly divided into 6 groups according to body weight: normal group [normal saline (NS) + NS]; model group (5 mg•kg^{-1} BLM + NS); HSYA groups (5 mg•kg^{-1} BLM + HSYA at 35.6, 53.3, and 80.0 mg•kg^{-1}), and DXM group (5 mg•kg^{-1} BLM + DXM 1 mg•kg^{-1}). Rats were anesthetized with pentobarbital intraperitoneally (i.p.). Then, rats in the BLM, HSYA, and DXM groups received an intratracheal delivery of BLM solution. Rats in the normal group received the same volume of 0.9% NaCl solution instead of BLM. The date of BLM administration was defined as day 0. From day 1, 0.9% NaCl was injected (i.p.) daily for 3 weeks to rats in the normal group and BLM group. Meanwhile, various doses of HSYA were injected (i.p.) daily to rats in HSYA groups and DXM was injected (i.p.) daily to rats of DXM group for 3 weeks. On day 7 or day 21, rats were anesthetized with pentobarbital sodium. Blood samples were obtained from the abdominal aorta with a heparinized needle for arterial blood gas analysis. Then, rats were sacrificed by exsanguination. Lung lobes were dissected and washed in ice-cold saline. Right lung lobes were snap-frozen in liquid nitrogen for RNA isolation, and Western blot analysis. Left lungs were weighed for the left lung index, and were then fixed in 10% neutral formalin for histopathological and immunohistochemical examination.

Arterial Blood Gas Analysis

Blood samples were obtained from the abdominal aorta and were measured using Radiometer ABL726 blood gas analyzer (Copenhagen, Denmark) at room temperature. Oxygen partial pressure (PaO_2), carbon dioxide partial pressure ($PaCO_2$), oxygen saturation (SO_2), pH, and HCO_3^- concentration of the samples were measured.

Histological Examination

Left lung samples were fixed in 10% neutral formalin, embedded in paraffin, and sectioned. Sections were stained with hematoxylin and eosin (HE) or Masson's trichrome according to conventional methods. The slices were evaluated under a light microscope.

Immunohistochemistry Analysis

Paraffin embedded tissue sections were rehydrated in xylene and graded ethanol solutions. The slides were exposed to 1% hydrogen peroxide/methanol for 10 min, washed in phosphate buffered saline (PBS), and then incubated with antigen retrieval buffer for 10 min at room temperature for antigen retrieval. After blocking with serum for 20 min, sections were then immunostained with α-SMA primary antibody or p65 antibody and incubated overnight at 4 ℃. After washing with PBS, sections were incubated with biotinylated secondary antibody for 20 min at room temperature. Sections were then washed with PBS and incubated for 10 min with a streptavidin-biotin-peroxidase complex. After washing with PBS, diaminobenzidine was added as a visualizing agent. Nuclei were counterstained with hematoxylin. Positive staining for α-SMA or p65 was brown. Expression of the target gene was compared between groups by calculating the number of positive staining cells. For each tissue section, the number of positively-stained cells was calculated from 10 different fields under a light microscope.

Reverse Transcription Polymerase Chain Reaction Analysis

Total RNA was isolated from each specimen of frozen lung tissue using TRIzol according to the manufacturer's instructions. RNA concentration and purity were determined by the Thermo Scientific NanoDrop 2000 (Wilmington, DE, USA). First-strand cDNA synthesis was performed with 2 μg of total RNA in a reaction volume of 20 μL using M-MLV reverse transcriptase and incubated at 42 ℃ for 1 h. One microliter of cDNA was amplified in 20 μL reaction using SYBR® Premix Ex Taq™ on an iCycler iQ real-time detection system. Polymerase chain reaction amplification was carried out as follows: initial denaturation at 95 ℃ for 30 s, 40 cycles with denaturation at 95 ℃ for 5 s, annealing at 60 ℃ for 30 s. The sequences of the primers was displayed in Table 1. Relative quantification was determined using the $2^{-\triangle\triangle Ct}$ method with data normalized to β-actin. Primers were produced by Beijing Sunbiotech Co., Ltd (Beijing, China).

Table 1. Sequence of Primers

mRNA	Sequence (5'–3')
TNF-α	F: GTGATCGGTCCCAACAAG
	R: AGGGTCTGGGCCATGGAA
IL-1β	F: CACCTCTCAAGCAGAGCACAGA
	R: ACGGGTTCCATGGTGAAGTC
IL-6	F: ATATGTTCTCAGGGAGATCTTGGAA
	R: GTGCATCATCGCTGTTCATACA
TGF-β1	F: CCCCTGGAAAGGGCTCAACAC
	R: TCCAACCCAGGTCCTTCCTAAAGTC
α-SMA	F: CGGGCTTTGCTGGTGATG
	R: GGTCAGGATCCCTCTCTTGCT
Collagen Ⅰ	F: AAAACGGGAGGGCGAGTG
	R: GGTCCCTCGACTCCTATGACTTC
β-actin	F: AGGCCAACCGTGAAAAGATG
	R: ACCAGAGGCATA

Hydroxyproline Assay of Lung Tissues

Hydroxyproline (HYP) content of lung was measured using HYP assay kits (Nanjing Jiancheng Biochemical Institute, Nanjing, China) according to the manufacturer's instructions.

Statistical Analysis

Data are presented as mean ± standard deviation ($\bar{x} \pm s$) and statistical analyses were performed by SPSS 16.0 software (SPSS Inc., Chicago, IL, USA). The results were analyzed by one-way analysis of variance (ANOVA) and SNK tests were used in *post hoc* multiple comparisons. Statistical significance was set at $P<0.05$.

RESULTS

Effect of HSYA on Left Lung Index of BLM-Induced Lung Injury

The left lung index was calculated to observe the severity of pulmonary edema. There was a significant increase in left lung index after BLM treatment on day 7 because of edema and on day 21 because of lung scarring with body weight loss. The HSYA and DMX groups had significantly lower left lung indices on both days 7 ($P<0.01$) and 21 (HSYA 53.3, 80.0 mg•kg^{-1},

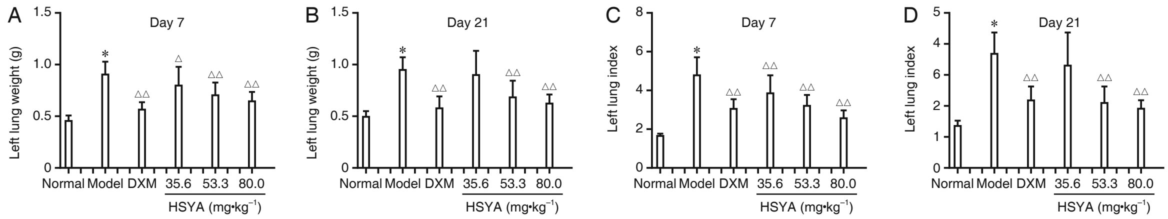

Figure 1. Effect of HSYA on Left Lung Weight and Left Lung Index in BLM-Induced Rats on Days 7 and 21 ($\bar{x} \pm s$)

Notes: (A) and (B): left lung weight; (C) and (D): left lung index. **P*<0.01 *vs*. the normal group; △*P*<0.05, △△*P*<0.01 *vs*. the model group; *n*=12 in each group

P<0.01, Figure 1).

Effect of HSYA on Changes in Arterial Blood PaO_2 and $PaCO_2$ Induced by BLM

Figure 2 shows that compared with the normal group, PaO_2 was significantly lower and $PaCO_2$ was significantly higher in the model group on both days 7 and 21 (*P*<0.01). Compared with the BLM group, the arterial blood PaO_2 in the DXM group was significantly higher and $PaCO_2$ significantly lower both on days 7 and 21 (all *P*<0.01). In the HSYA groups, arterial blood PaO_2 was significantly higher and $PaCO_2$ was significantly lower, especially in the 80.0 mg•kg^{-1} HSYA group (*P*<0.01), both on days 7 and 21, compared with the model group.

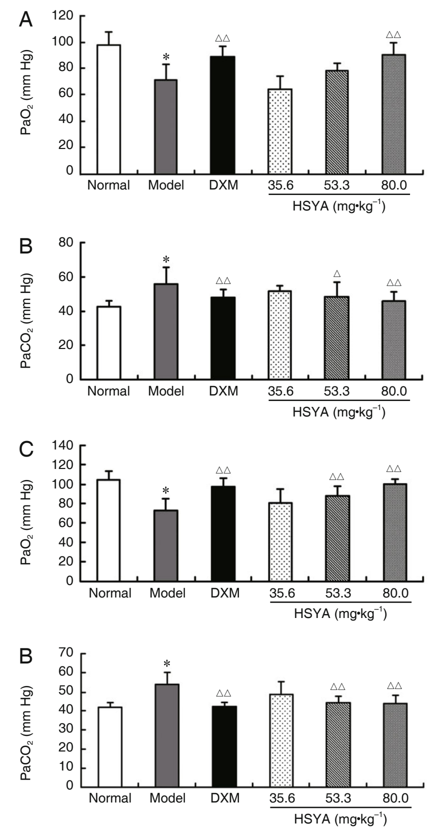

Figure 2. Effect of HSYA on Changes in Arterial Blood PaO_2 and $PaCO_2$ Induced by BLM ($\bar{x} \pm s$)

Notes: **P*<0.01, *vs*. the normal group; △*P*<0.05, △△*P*<0.01, *vs*. the model group. (A) PaO_2 on day 7; (B) $PaCO_2$ on day 7; (C)PaO_2 on day 21; (D) $PaCO_2$ on day 21; *n*=8 in each group.

Effect of HSYA on BLM-Induced Pulmonary Morphological Changes

Figure 3A shows that on day 7, the alveoli in the normal group were clearly visible and their structure was intact with no inflammatory edema. The model group had collapsed and shrunken alveoli, and the matrix was obviously thicker with serious inflammatory cell infiltration and edema. The DXM group had mostly intact alveoli with slight inflammatory cell infiltration and edema. The 35.6 mg•kg^{-1} HSYA group had mostly collapsed alveoli, with a few left intact. The matrix was thick and moderate tissue edema was obvious. The 53.3 mg•kg^{-1} HSYA group had intact alveolus structure and the edema and the other inflammatory injury was less serious than that in the 35.6 mg•kg^{-1} HSYA group. The 80.0 mg•kg^{-1} HSYA group had a clear alveolus structure with slight matrix thickening and inflammatory cell infiltration; there was no significant edema.

Figure 3B shows the lungs of the normal group on day 21. The structure of the alveoli were intact with no matrix thickening or inflammatory cell infiltration. The model group had obvious focal consolidation and significant matrix thickening and inflammatory cell infiltration. The DXM group had mostly intact alveolus structure with slight inflammatory injury and no focal consolidation or pulmonary alveolus wall thickening. In the low-dose HSYA group, there was significant thickening in the alveolar wall and inflammatory injury with consolidation in many regions. In the middle-dose HSYA group, the alveoli were intact and the focal consolidation was smaller than that in the low-dose HSYA group. In the high-dose HSYA group, the structure of most alveoli were intact with no thickening of the alveolar wall, consolidation, or significant inflammatory cell infiltration.

Figure 3C shows that on day 21 in the model

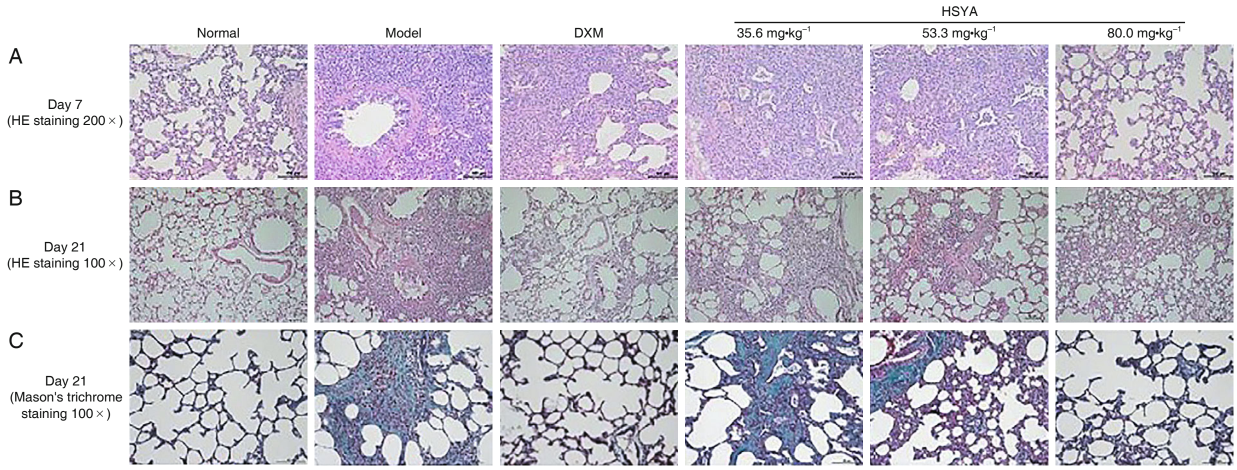

Figure 3. Effect of HSYA on Lung Histopathology Induced by BLM

Notes: (A) Day 7 HE staining: The injuries were attenuated in animals treated by HSYA or DXM. (B) Day 21 HE staining 100×: the injuries were attenuated in animals treated by HSYA and DXM. (C) Day 21 Masson's trichrome staining 200×: treatment with HSYA alleviated collagen deposition induced by BLM, and the effect was more significant with 80.0 mg•kg^{-1} HSYA.

group, there was blue-stained collagen by Masson's trichrome staining, especially in the areas with focal consolidation.

Figure 4 shows that the lung HYP level was higher in the model group than that in the normal group (P<0.01). DXM and 80 mg•kg^{-1} HSYA treatment significantly attenuated these increases (P<0.05).

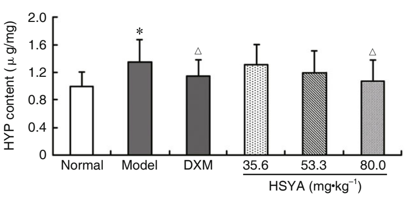

Figure 4. Effect of HSYA on HYP Content of Lung Tissue

Notes: $^{*}P$<0.01, *vs*. the normal group; $^{\triangle}P$<0.05, *vs*. the model group; n=12 in each group

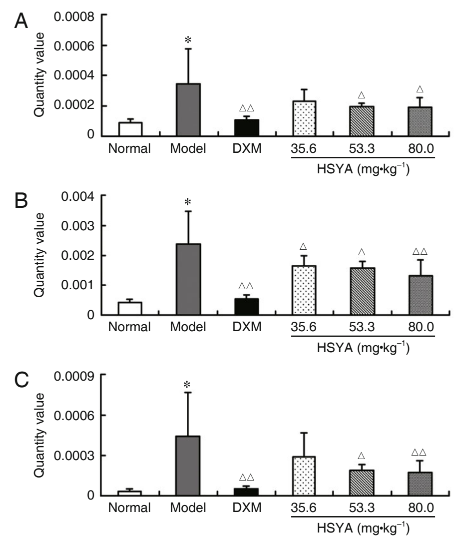

Figure 5. Effect of HSYA on mRNA Expression of TNF-α (A), IL-1β (B), and IL-6 (C) on Day 7

Notes: $^{*}P$<0.01, *vs*. with the normal group; $^{\triangle}P$< 0.05, $^{\triangle\triangle}P$<0.01 *vs*. the model group; n=8 in each group

Effects of HSYA on BLM-Induced TNF-α, IL-1β, and IL-6 mRNA Expression

The mRNA levels of TNF-α, IL-1β, and IL-6 in the lung were significantly higher in the model group compared with the normal group (P<0.01). This high expression was attenuated by DXM and dose-dependently by HSYA (53.3, 80.0 mg•kg^{-1}) on day 7 (P<0.05 or P<0.01, Figure 5).

Inhibitory Effect of HSYA on Lung NF-κB p65 Level on Day 7

To observe the anti-inflammatory effects of HSYA, the level of lung p65 subunit was observed by immunohistochemistry on day 7. As shown in Figure 6, the lung p65 positive cell number was significantly higher, especially in areas of inflammation, on day 7 after BLM administration compared with the normal group. Cells positive for p65 were rare in lungs of rats of normal group. Treatment with HSYA inhibited the BLM-induced increase in p65 positive cells, and this effect was more apparent in the 80.0 mg•kg^{-1} HSYA group (53.3 mg•kg^{-1}, P<0.05; 80.0 mg•kg^{-1}, P<0.01). Treatment with DXM also significantly attenuated this change (P<0.01).

Effect of HSYA on BLM-Induced Pulmonary TGF-β1, α-SMA, and Collagen Ⅰ mRNA Levels and α-SMA Protein Levels

Figure 7 shows that the levels of TGF-β1,

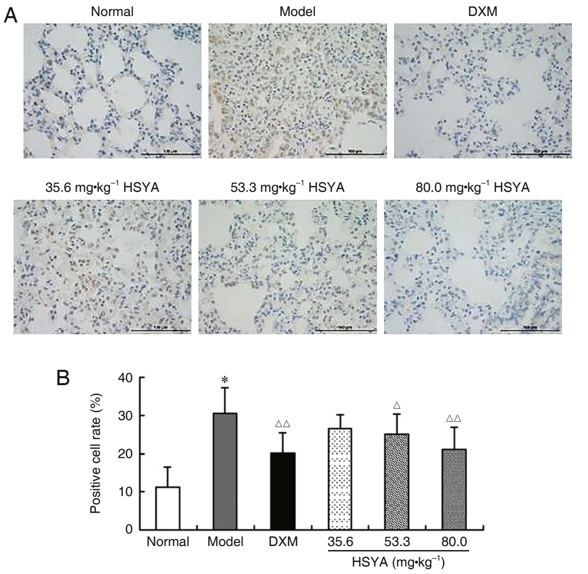

Figure 6. Effect of HSYA on Lung NF-κB p65 Level on Day 7 (400×)

Notes: Immunohistochemical staining pictures (A) and the analysis results of scanning data (B). *P<0.01, *vs*. the normal group; △P<0.05, △△P<0.01, *vs*. the model group; n=8 in each group

α-SMA, and collagen Ⅰ mRNA were higher in the model group compared with the normal group on day 21. HSYA (53.3, 80.0 mg•kg^{-1}) or DXM groups had significantly lower levels than those in the model group (P<0.05 or P<0.01).

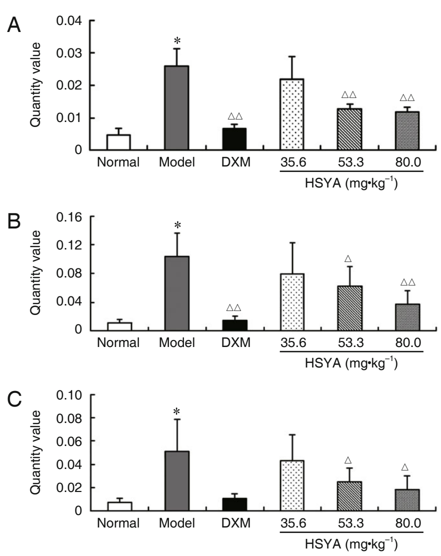

Figure 7. Inhibition of HSYA on BLM-Induced Pulmonary TGF-β1 (A), α-SMA (B), or Collagen Ⅰ (C) mRNA Elevation in Rats on Day 21

Notes: *P<0.01, *vs*. normal group; △P<0.05, △△P<0.01, *vs*. model group; n=8 in each group

After lung fibroblasts differentiate into myofibroblasts, α-SMA expression increases. The normal group has only a few α-SMA positive cells in the peribronchial region. In the model group, the number of positive cells was significantly higher than that in the normal group, especially in fibrous areas. The HSYA group had significantly fewer α-SMA positive cells compared with the model group. There were fewer α-SMA positive cells in the 80.0 mg•kg^{-1} HSYA group than other HSYA groups (Figure 8).

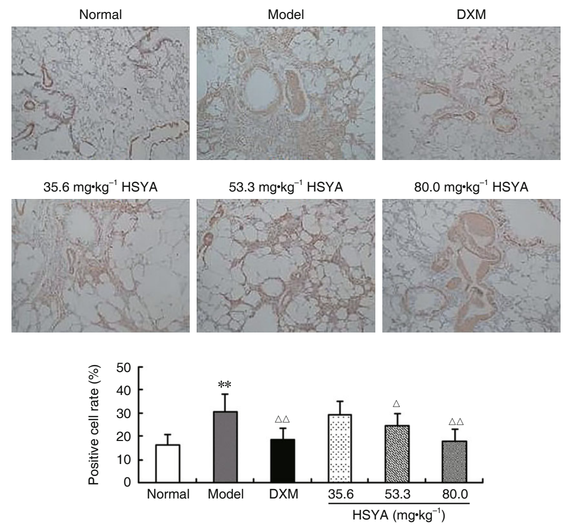

Figure 8. Inhibition of HSYA on BLM-Induced Pulmonary α-SMA Immunohistochemical Staining Elevation (100×)

Notes: Immunohistochemical staining pictures (A) and the analysis results of scanning data (B). *P<0.01, *vs*. normal group; △P<0.05, △△P<0.01, *vs*. model group; n=8 in each group

DISCUSSION

In this study, we demonstrated the beneficial effects of HSYA on early inflammation and late fibrosis in BLM-induced rat pulmonary injury. BLM exposure resulted in pulmonary edema and significant leukocyte infiltration in the lung on day 7. Arterial blood gas analysis showed hypoxemia and acidosis, which indicated lung respiratory dysfunction. HSYA ameliorated arterial gas changes dose-dependently. As a result of BLM-induced injury, rats experienced a marked body weight loss and lung index increase. Intraperitoneal injection of HSYA reduced the lung index and inflammatory cell infiltration. The anti-inflammatory agent DXM also attenuated the inflammatory response by reducing leukocyte infiltration and ameliorating gas exchange impairment. However, the loss of body weight was more severe after DXM treatment. This result may be correlated with its side effects of inhibiting protein synthesis and promoting protein decomposition. HSYA significantly attenuated the body weight loss, especially at a dose of 80.0 mg•kg^{-1}. These results demonstrated the

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protective effect of HSYA on inflammation and fibrosis in BLM-induced rat lung injury.

BLM is a chemotherapeutic antibiotic produced from *Streptomyces verticillus*. Previous study showed that toll-like receptor (TLR) 2 recognizes BLM as a pathogen-associated molecule. This recognition triggers the activation of intracellular signaling pathways involving transcription factors such as NF-κB and MAPKs that lead to the upregulation of genes encoding chemokines and proinflammatory cytokines such as TNF-α, IL-1β, and IL-6.[11,12] Another study demonstrated that TLR2 translated oxidative tissue damage into inflammatory responses by mediating the effects of oxidized phospholipids. HSYA was found to have the effect of attenuate inflammatory injuries, and antioxidation may be one of the mechanisms of inhibiting these injuries.[13] Another study found TLR2-dependent inflammatory gene expression and JNK MAPK and p38 MAPK signaling in macrophages.[14] In the first stage of BLM-induced lung injury, there is increased expression of many inflammatory cytokines in the lung that are accompanied by recruitment of inflammatory cells and cytokines. These cytokines regulate the inflammatory process, which is related to the accumulation of extracellular matrix components in fibrosis development.

Previous studies showed that the expression of TNF-α, IL-1β, and IL-6 were up-regulated in both BLM-induced animals and idiopathic pulmonary fibrosis patients.[15,16] These cytokines are related to the recruitment of inflammatory cells and induction of the pro-fibrotic cytokine TGF-β1.[17,18] We found that BLM administration induced a significant increase in TNF-α, IL-1β, and IL-6 mRNA expression via RT-PCR. The increase in these proinflammatory cytokines was significantly attenuated by treatment with HSYA. During lung injury, NF-κB is activated, which results in gene expression of some cytokines, chemokines, and growth factors.[19] We further assessed the level of the NF-κB p65 subunit with immunohistochemistry. The level of p65 subunit was higher after BLM administration compared with normal animals. Treatment with HSYA significantly attenuated the increased p65 positive cell rate. These results agree with our previous study using a lipopolysaccharide-induced acute lung injury mouse model. Recently, HSYA was also found to alleviate CCl_4-induced liver fibrosis in rats.[5]

The mRNA levels of the pro-fibrogenic cytokine TGF-β1 were also evaluated in this study. As a link between inflammation and fibrosis, TGF-β1 is a key cytokine in fibrogenesis.[20] TGF-β1 promotes the proliferation and differentiation of fibroblasts into activated myofibroblasts, and also promotes the epithelial mesenchymal transition, which is another source of myofibroblasts. At the same time TGF-β1 enhances collagen and fibronectin production and reduces extracellular matrix degradation.[21] We found that HSYA attenuated the increased mRNA level of TGF-β1 induced by BLM. This regulation of TGF-β1 might be partly explained by the downregulation of TNF-α, IL-1β, and IL-6.

We found that HSYA could significantly alleviate BLM-induced early pulmonary inflammation and late pulmonary fibrosis by suppressing the activation of NF-κB and α-SMA. Our data suggest that HSYA might be effective at attenuating pulmonary fibrosis. HSYA is highly water-soluble and is not cell membrane-permeable.[22] We hypothesize that HSYA might target the cell membrane and interfere with ligands (e.g. microbial ligands, proinflammatory cytokines, or growth factors) by binding to their receptors and exerting its effects through downstream signal transductions. However, the mechanism by which HSYA alters intracellular signaling requires further study.

In rat model BLM triggered significant pulmonary edema, gas exchange disturbance and pulmonary bullation could be observed just like changes in interstitial pneumonia patient. These changes could be attenuated by HSYA, therefore, it suggested that HSYA might be effective in attenuating the interstitial pneumonia. HSYA is active to attenuate BLM-induced early inflammation[8] and in this report we found that HSYA in the same dose could inhibit both inflammation and fibrosis in this model. BLM induced pulmonary fibrosis is reversible but the fibrosis is irreversible in idiopathic pulmonary fibrosis patients[3], so BLM induced pulmonary fibrosis is different from the IPF patient in some respect. The effect of HSYA to treat pulmonary fibrosis need to be proved by further clinic trial.

HSYA alleviated acute lung inflammation and chronic pulmonary fibrosis induced by BLM, and inhibited the activation of lung fibroblasts. These results suggest that the anti-fibrotic effect of HSYA may be associated with the inhibition of lung inflammation and lung fibroblast activation.

Conflicts of Interest

None.

Author Contributions

Jing M was responsible for the experimental design and wrote the manuscript; Wang L performed most of the animal experiments; Wu Y and Tan L performed some of the animal experiments; Zang BX performed the HSYA HPLC analysis.

REFERENCES

1. Fan L, Zhao HY, Xu M, Zhou L, Guo H, Han J, et al. Qualitative evaluation and quantitative determination of 10 major active components in *Carthamus tinctorius* L. by high-performance liquid chromatography coupled with diode array detector. J Chromatogr A 2009;1216:2063-2070.
2. Zhang Y, Guo J, Dong H, Zhao X, Zhou L, Li X, et al. Hydroxysafflor yellow A protects against chronic carbon tetrachloride-induced liver fibrosis. Eur J Pharmacol 2011;660:438-444.
3. Moeller A, Askc K, Warburton D, Gauldie J, Kolb M. The bleomycin animal model: A useful tool to investigate treatment options for idiopathic pulmonary fibrosis? Int J Biochem Cell Biol 2008;40:362-382.
4. Matute-Bello G, Frevert CW, Martin TR. Animal models of acute lung injury. Am J Physiol Lung Cell Mol Physio 2008;295:L379-L399.
5. Chaudhary NI, Schnapp A, Park JE. Pharmacologic differentiation of inflammation and fibrosis in the rat BLM model. Am J Respir Crit Care Med 2006;173:769-776.
6. Bringardner BD, Baran CP, Eubank TD, Marsh CB. The role of inflammation in the pathogenesis of idiopathic pulmonary fibrosis. Antioxid Redox Signal 2008;10:287-301.
7. Sun CY, Pei CQ, Zang BX, Wang L, Jin M. The ability of hydroxysafflor yellow A to attenuate lipopolysaccharide-induced pulmonary inflammatory injury in mice. Phytother Res 2010;24:1788-1795.
8. Wu Y, Wang L, Jin M, Zang BX. Hydroxysafflor yellow A alleviates early inflammatory response of bleomycin-induced mice lung injury. Biol Pharm Bull 2012;35:515-522.
9. Wang L, Jin M, Zang BX, Wu Y. Inhibitory effect of safflor yellow on pulmonary fibrosis. Biol Pharm Bull 2011;34:511-516.
10. Zang BX, Jin M, Li JR. Large scale preparation of pure hydroxysafflor yellow A by macroporus resin-gel column chromatography. J Cardiovasc Pulm Dis 2008;27:363-365.
11. Razonable RR, Henault M, Paya CV. Stimulation of toll-like receptor 2 with bleomycin results in cellular activation and secretion of pro-inflammatory cytokines and chemokines. Toxicol Appl Pharmacol 2006;210:181-189.
12. Yang HZ, Cui B, Liu HZ, Chen ZR, Yan HM, Hua F, et al. Targeting TLR2 attenuates pulmonary inflammation and fibrosis by reversion of suppressive immune microenvironment. J Immunol 2009;182:692-702.
13. Wang XN, Xu XM, Qiu LH, Li G, Xu B, Ma CJ, et al. Effect of hydroxysafflor yellow A on the oxidation of intracellular glutathione induced by hydrogen peroxide in L02 cells. Food Sci (Chin) 2013;34:317-320.
14. Kadl A, Sharma PR, Chen W, Agrawal R, Meher AK, Rudraiah S, et al. Oxidized phospholipid-induced inflammation is mediated by Toll-like receptor 2. Free Radic Biol Med 2011;51:1903-1909.
15. Pan LH, Ohtani H, Yamauchi K, Nagura H. Co-expression of TNF alpha and IL-1 beta in human acute pulmonary fibrotic diseases: an immunohistochemical analysis. Pathol Int 1996;46:91-99.
16. Zhang Y, Lee TC, Guillemin B, Yu MC, Rom WN. Enhanced IL-1 beta and tumor necrosis factor-alpha release and messenger RNA expression in macrophages from idiopathic pulmonary fibrosis or after asbestos exposure. J Immunol 1993;150:4188-4196.
17. Sime PJ, Marr RA, Gauldie D, Xing Z, Hewlett BR, Graham FL, et al. Transfer of tumor necrosis factor-alpha to rat lung induces severe pulmonary inflammation and patchy interstitial fibrogenesis with induction of transforming growth factor-beta1 and myofibroblasts. Am J Pathol 1998;153:825-832.
18. Kolb M, Margetts PJ, Anthony DC, Pitossi F, Gauldie J. Transient expression of IL-1beta induces acute lung injury and chronic repair leading to pulmonary fibrosis. J Clin Invest 2001;107:1529-1536.
19. Christman JW, Sadikot RT, Blackwell TS. The role of nuclear factor-kappa B in pulmonary diseases. Chest 2000;117:1482-1487.
20. Bonniaud P, Margetts PJ, Ask K, Flanders K, Gauldie J, Kolb M. TGF-beta and Smad3 signaling link inflammation to chronic fibrogenesis. J Immunol 2005;175:5390-5395.
21. Coward WR, Saini G, Jenkins G. The pathogenesis of idiopathic pulmonary fibrosis. Ther Adv Respir Dis 2010;4:367-388.
22. Ma GN, Yu FL, Zhang H, Li ZP, Mei XG. Pharmacokinetics and biopharmaceutics of hydroxysafflor yellow A: research advances. J Int Pharm Res 2014;41:195-199, 220.
23. Zhang Q, Chen ZY, Wu LB, Wang SC, Zheng QS. Clinical non-feriority evaluation on the efficacy and safety of safflower yellow pigment lyophilized powder and dripping solution in the treatment of patients with angina. Chin J Evid-Based Med (Chin) 2005;15:276-285.
24. Pei CQ, Sun CY, Jin M. Mitigation of safflor yellow injection on acute lung injury of rats induced by oleic acid. Chin Tradit Herb Drugs (Chin) 2010;41:597-601.

(Received June 3, 2015)
Edited by WANG Wei-xia